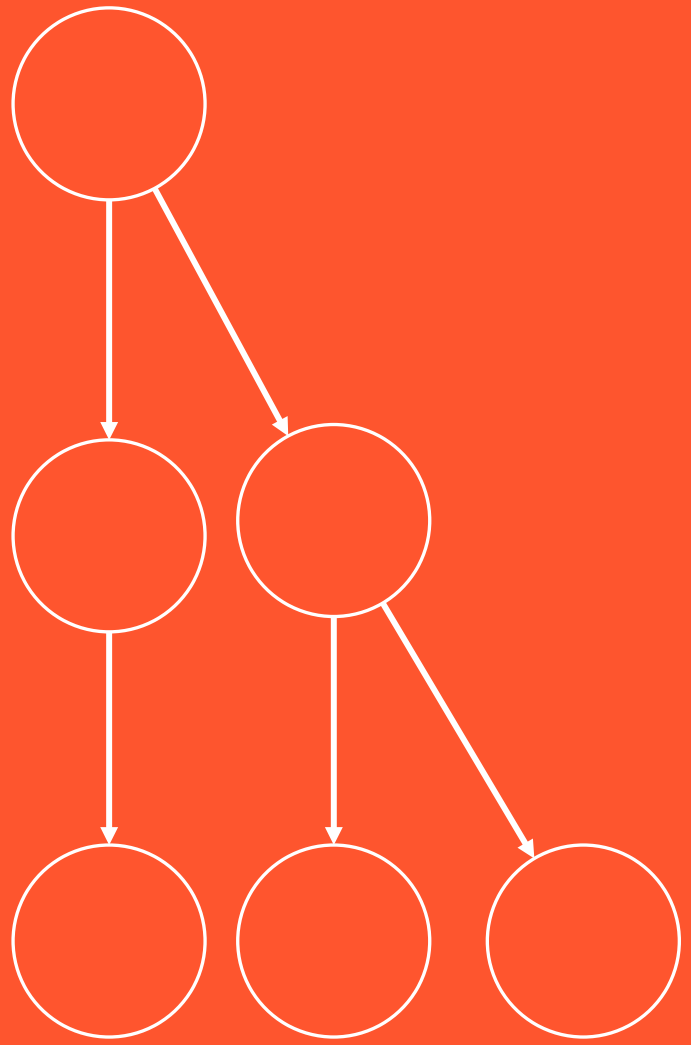


BCOG 100

Memory, Imagination, and Generative Models

Lecture 10.2 — Storing, Reconstructing, and Simulating

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: A Door Code

In your group, three minutes.

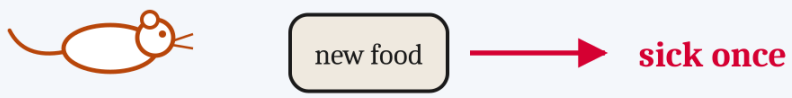
What main problem did we describe in the last lecture, in one sentence?

Then: you learn a door code so that you can get into a building tomorrow. Besides opening that door, what else could you do with the code?

A Disposition, and Accessible Content

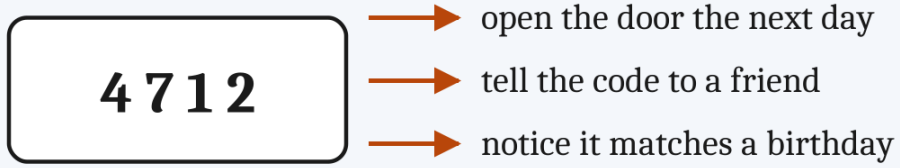
A **disposition** is a learned change in a tendency to respond. **Accessible content** is information that can be accessed later and combined with other information when a situation requires it.

A disposition



A learned change in a tendency to respond: the rat avoids that food afterward. With only that tendency, it has no information about where or when it ate the food.

Accessible content



Information that can be accessed later and combined with other information when a situation requires it. The two kinds can coexist.

Where the line is drawn

Not by whether the use was settled when it was learned: the door code had one known use, and it is still accessible content.
By whether the information can be accessed and combined with other information. This revises the line we drew in Module 5.

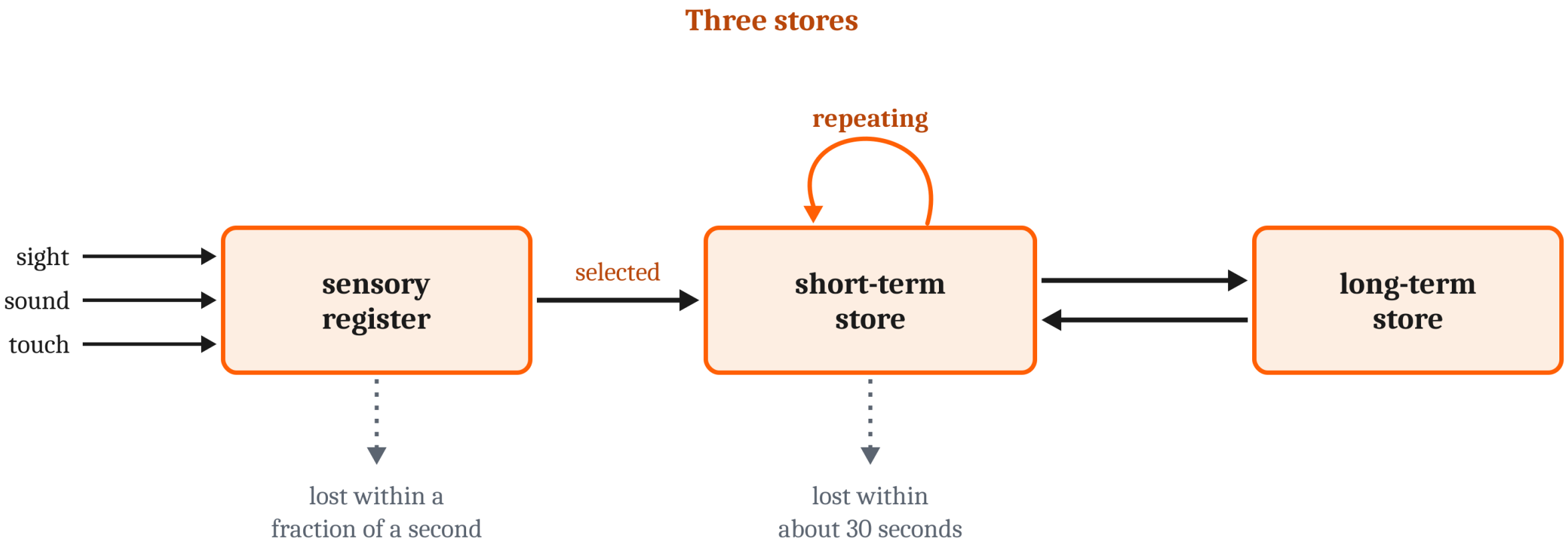
Kinds of Memory

Three kinds by how long, then long-term memory divided by what it contains.

Kind of memory	How long	An example
Sensory memory	About a second	The rustle the early mammal just heard
Working memory	Seconds to minutes in people; in some animal research, one occasion	Digits repeated back; the maze arms a rat has entered
Long-term memory	Days, months, or years	Procedural, semantic, and episodic memory
Procedural memory	Days to a lifetime	Running a familiar route
Semantic memory	Days to a lifetime	Beetles are usually found under loose bark
Episodic memory	Days to a lifetime	Where a person parked on one particular day
Episodic-like memory	Up to twelve days, in the tests from the first lecture	Scrub jays searching first for worms after four hours, and for peanuts after five days

A Store That Fades, and a Few Items at a Time

Rehearsal: repeating items to oneself so they are not lost. A **chunk** is a group of items a person already knows as one unit.



after Atkinson and Shiffrin (1968), Fig. 1; times are the authors' estimates (pp. 90–95)

What Tulving Meant, and Can Animals Remember an Occasion?

Episodic-like memory: memory for what happened, where, and when, on one occasion, as shown by behavior alone.

Tulving, 1972

Episodic memory named, against semantic memory: called "systems" for convenience, not to claim they were separate.

The self from the start: "I did such and such, in such and such a place, at such and such a time."

Tulving, 1985

The defining feature: re-living an occasion as part of one's own past (autonoetic, or "self-knowing").

N.N. knew the year his family moved and the names of his schools, but recalled no event. Tulving classed that knowledge as semantic.

Animals: behavior alone

Episodic-like memory: what happened, where, and when, on one occasion, shown by behavior alone.

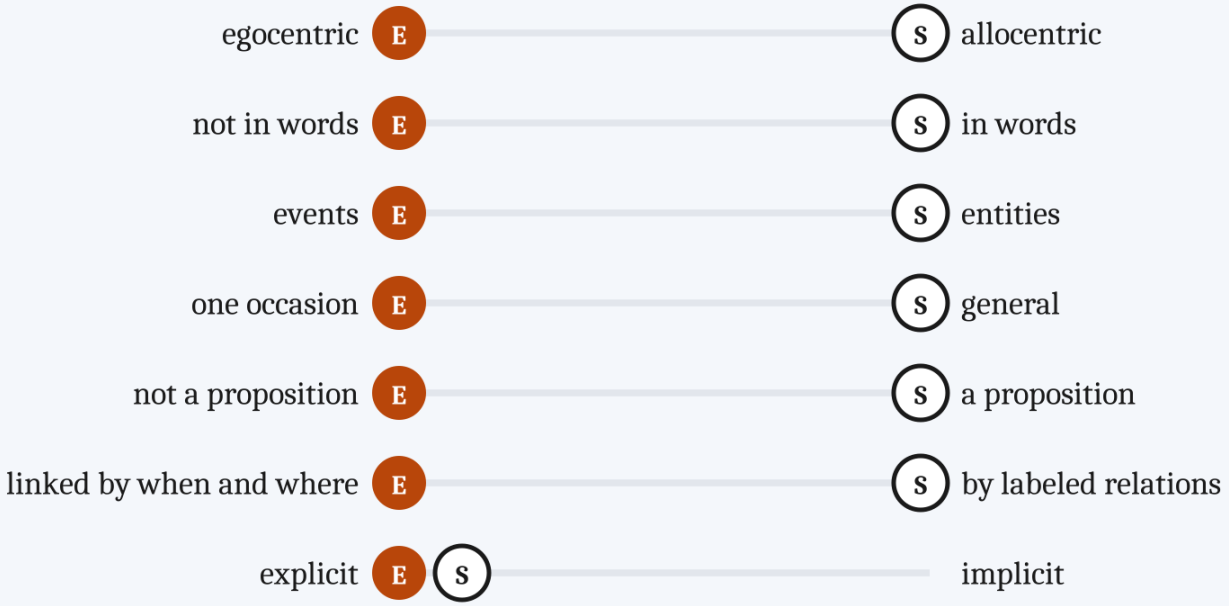
Several species pass such tests. Re-living is "probably undetectable in many species". Researchers disagree about what follows.

A video recorder records events, yet "No one would claim that a video recorder has episodic memory".

Knowing Without Being Able to Say, and Bundles of Dimensions

Declarative memory is available to consciousness and can be expressed by language. **Nondeclarative memory** is expressed in performance rather than recollection.

One way to lay out how memories vary



E, typical episodic memory; S, typical semantic memory. Each dimension has a published case that does not fit. Missing from the seven: being about oneself (personal semantics).

Is it accessible content?

Inferred from several kinds of evidence together:

- report, or other performance that depends on the content
- use outside the place or context of its initial presentation
- a selective effect of an intervention, with perception, motivation, and movement ruled out

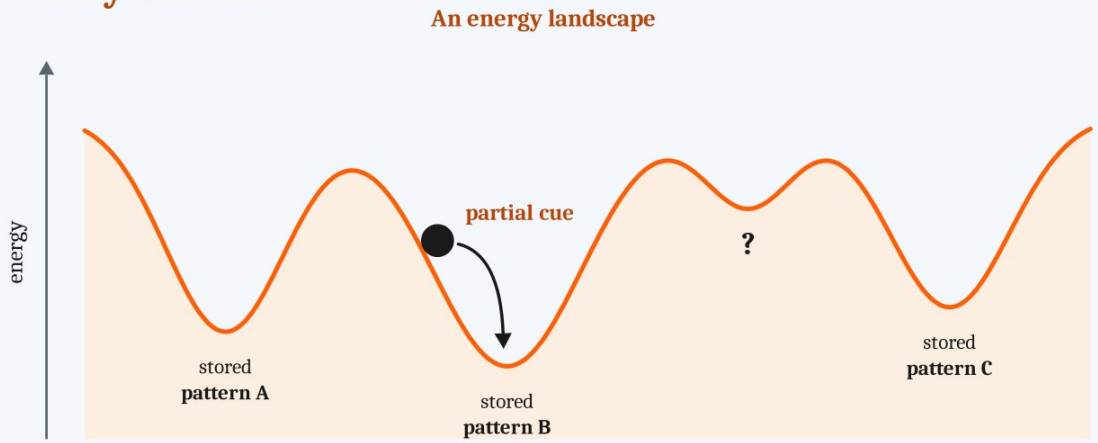
K.C.

Recalled no occasion of his life, yet knew technical terms learned in the same years.

Recall by Content, and Forgetting

A **content-addressable memory** finds a stored item from part of the item itself. **Interference** is the effect of other learning on recalling something.

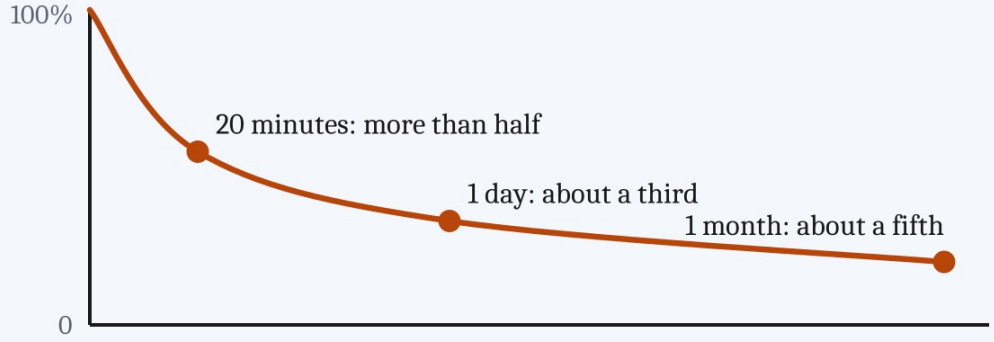
Recall by content



schematic, after Hopfield (1982); each point along the line is one state of the whole network

Settles into the nearest stable state. Capacity: about 15 patterns in 100 units. Past it, new patterns make every stored one irretrievable.

Forgetting: fast, then slower



Ebbinghaus: time saved in relearning

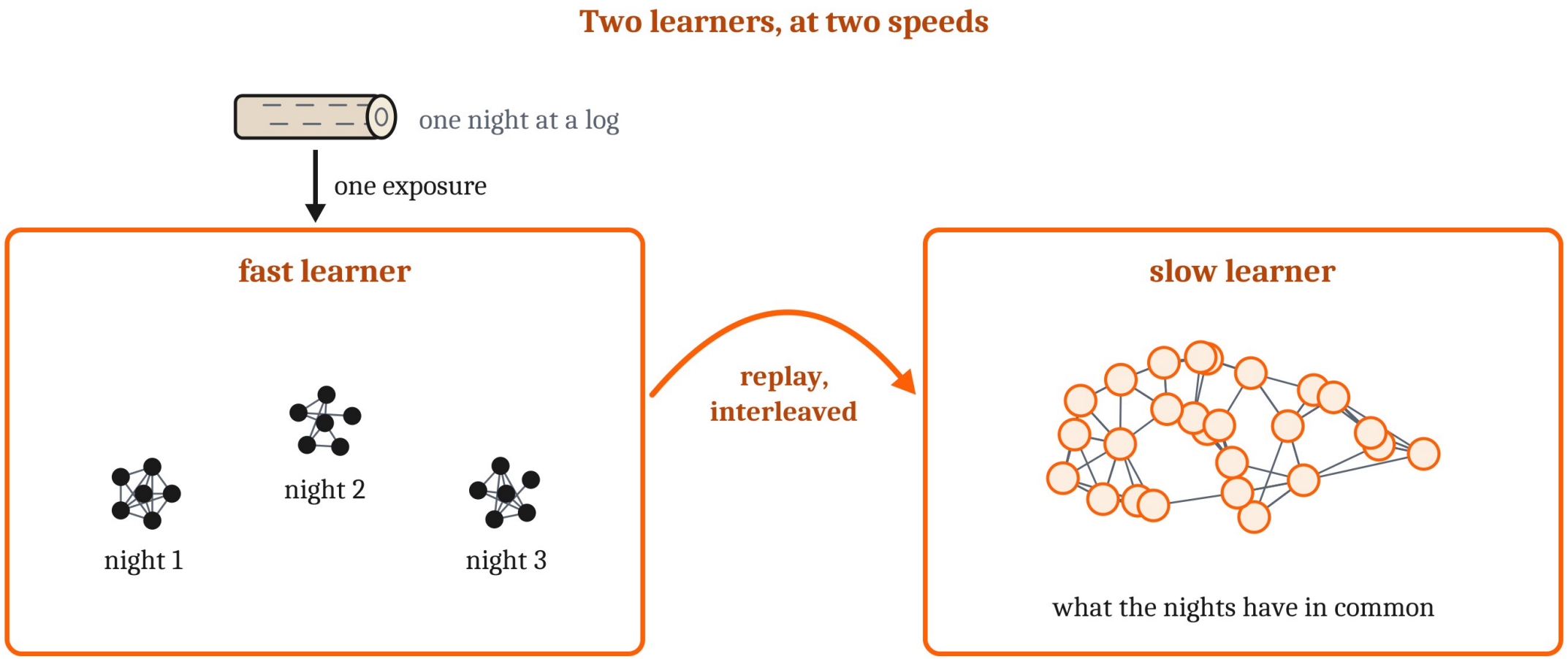
Catastrophic interference

After a second list giving the same first items new partners, first pairs still recalled:

people	more than half
a standard network	none

Two Learners at Two Speeds, and Machines That Learn One Thing After Another

Complementary learning systems: a slow learner of shared structure, and a fast learner of single occasions that replays those occasions to the slow learner.



schematic, after McClelland, McNaughton, and O'Reilly (1995) and O'Reilly and colleagues (2012)

Remembering as Reconstruction

Reconstructive memory: remembering rebuilds an occasion from general knowledge and a few specifics, rather than reproducing a copy.

Read the list once, at a steady pace.

bed	rest	awake	tired	dream
wake	night	blanket	doze	slumber
snore	pillow	peace	yawn	drowsy

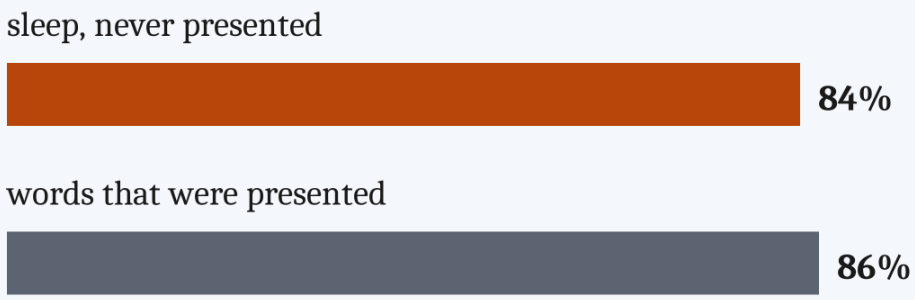
Then look away, and write down as many as you can.

Remembering as Reconstruction

Reconstructive memory: remembering rebuilds an occasion from general knowledge and a few specifics, rather than reproducing a copy.

Was sleep on the list?

No, though every word on it is associated with sleep. On a later test, the share judged to have been on the list:



Most who said sleep was on the list reported remembering hearing it.

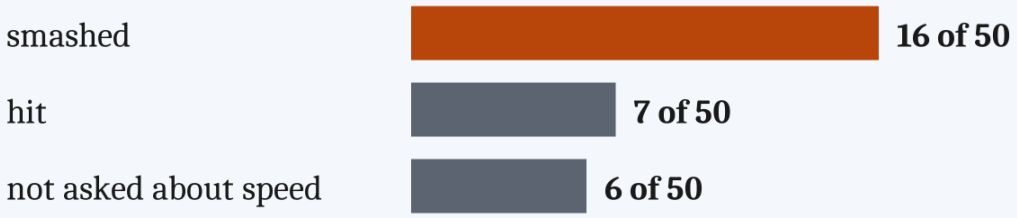
One word in a question

How fast were the cars going when they "smashed" or "hit" each other?

Estimated speed, miles per hour

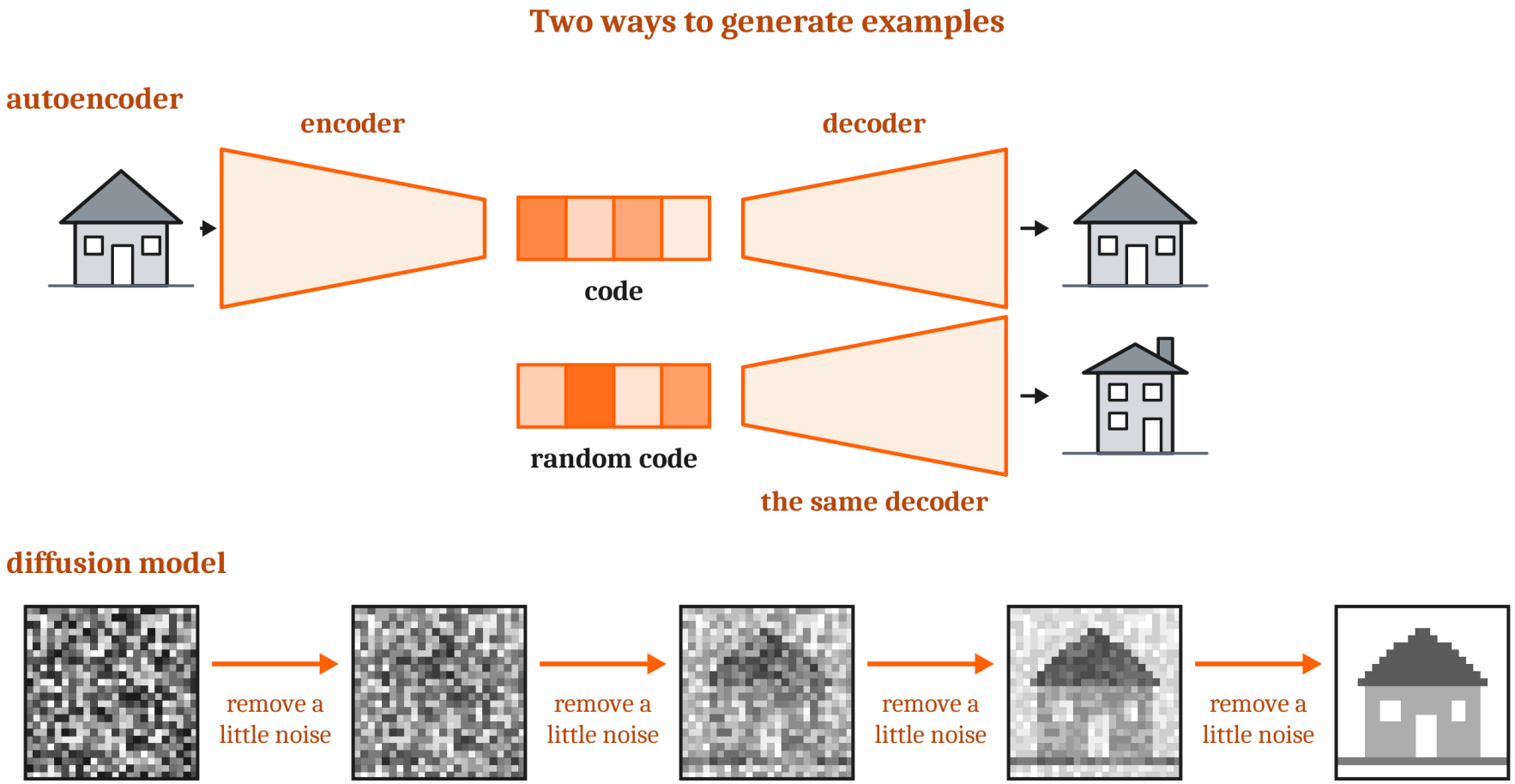


A week later, saw broken glass (none in the film)



Generative Models, and What They Reproduce

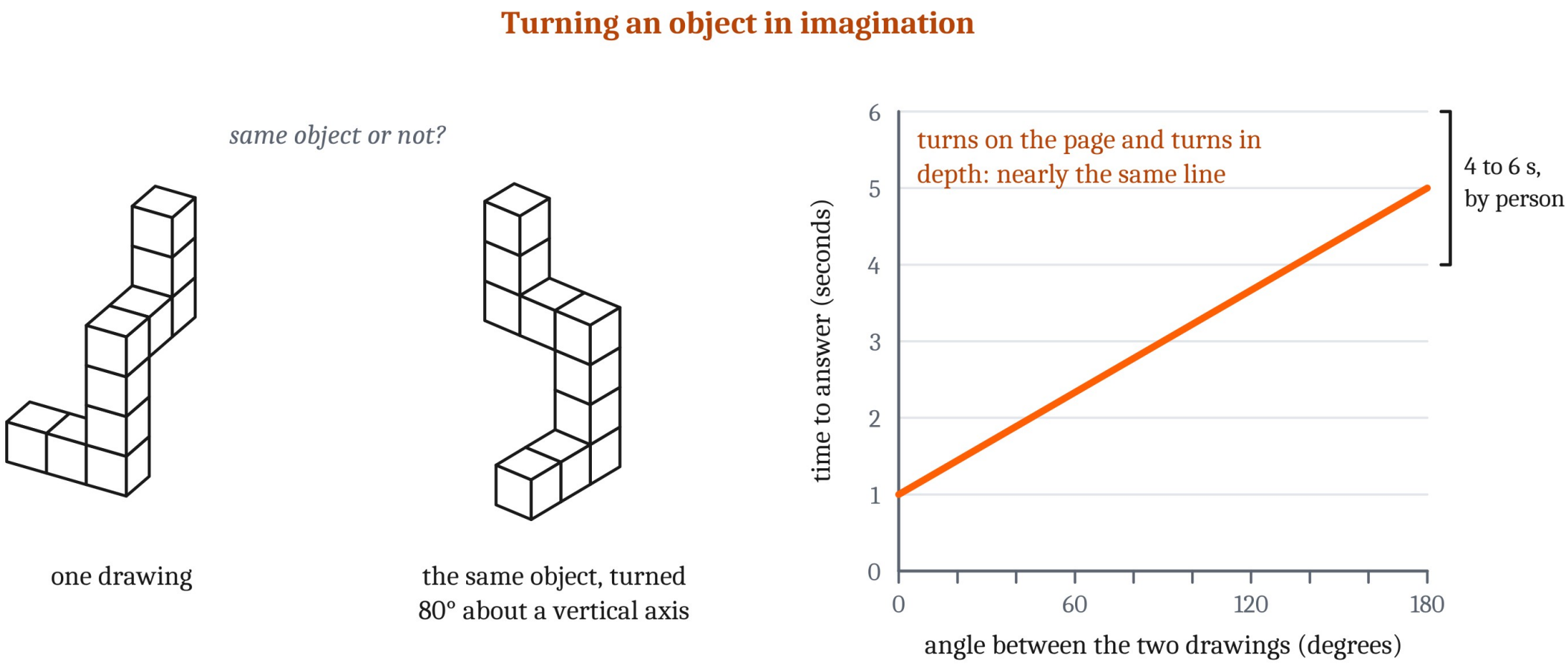
A **generative model** learns from many examples what kinds of things occur, and can produce new examples of the same kinds.



schematic, after Prince (2026), chapters 17 and 18

Mental Images

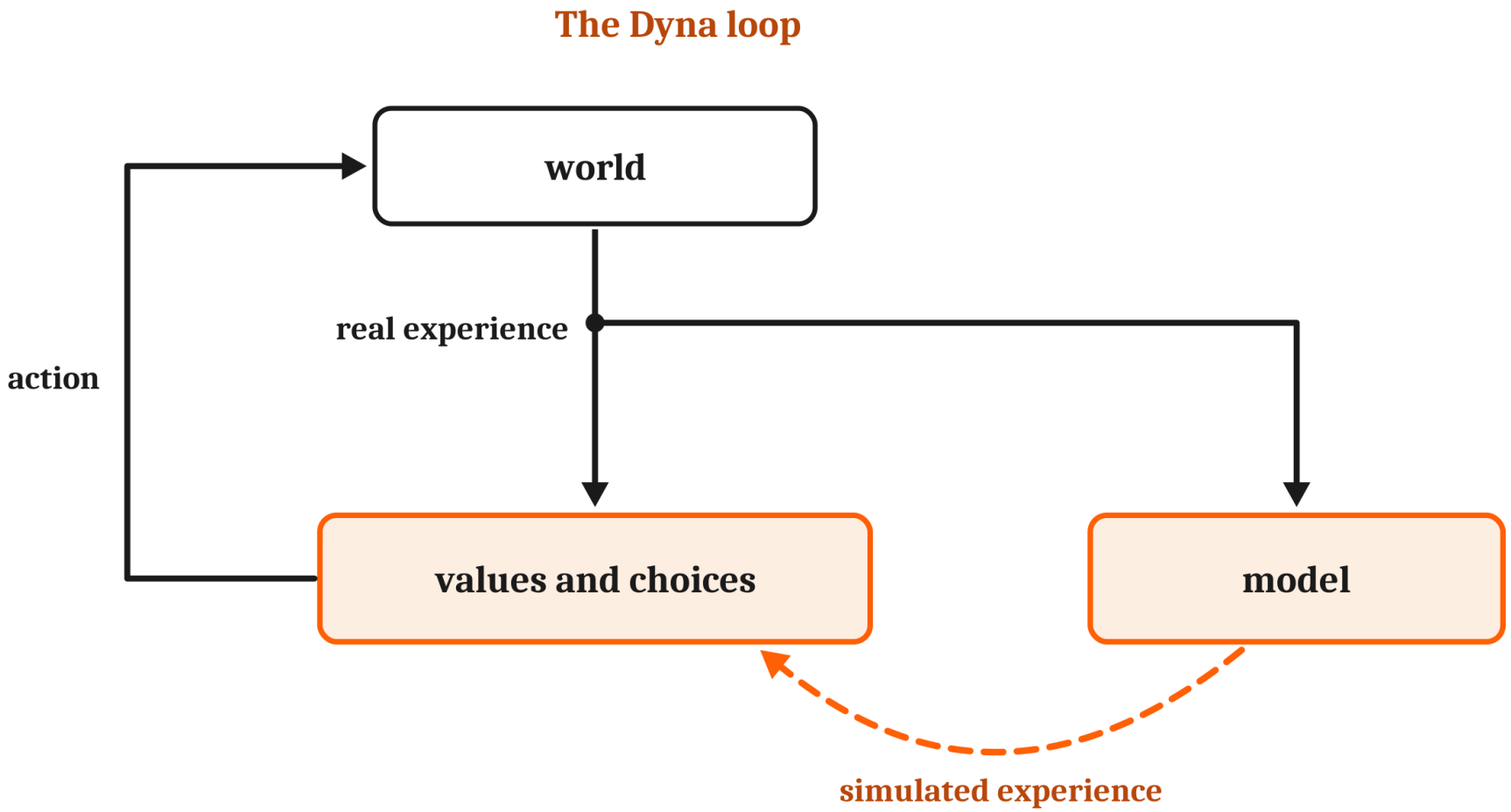
Mental imagery is an experience resembling perception, of something not present to the senses. Judging a turned object took longer the farther it was turned.



line drawn from the reported relation, not from the original data points; after Shepard and Metzler (1971), Figs. 1 and 2

Learning from What Did Not Happen

Simulation is using stored knowledge, without acting, to predict what would happen. In Dyna, simulated experience trains the same values as real experience.



schematic, after Sutton (1991), Fig. 2; real and simulated experience train the values by the same rule

To Discuss: What Kind of Memory Does a Chatbot Have?

In your group, five minutes.

A chatbot answers questions about general facts, and about something said earlier in the same conversation.

Using the terms from this lecture: what kinds of memory does it have? Is what it has a disposition, accessible content, or both? Does it reconstruct, the way people do?

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